

This means that the velocity of the vehicle is 100 m s^{-1} .

An examination of the triangle CDE shows that this graph has a gradient given by $\frac{DE}{CE}$, which gives a velocity of $2 \times 10^2 \text{ m s}^{-1}$.

If the velocity-time graph of the slower vehicle was plotted, another useful aspect of graphs can be illustrated, as shown in Figure 3.12.

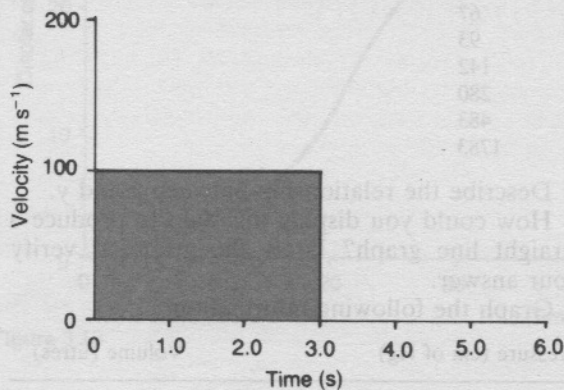


Figure 3.12

The area under the curve at any given time is the product of velocity and time, which is displacement. For the displacement after 3.0 s the area under the curve is determined as follows.

$$\begin{aligned}\text{area under curve} &= \text{area of rectangle} \\ &= \text{time} \times \text{velocity} \\ &= 3.0 \text{ s} \times 100 \text{ m s}^{-1} \\ \therefore \text{displacement} &= 300 \text{ m}\end{aligned}$$

This means that the distance travelled in 3.0 s is 300 m.

A further example of the analysis of graphs can be seen when a velocity-time graph is drawn for a uniformly accelerating vehicle. The graph produced is a straight line of best fit, originating at origin if the vehicle starts from rest. This can be seen in Figure 3.13.

The gradient of the line will provide the acceleration of the vehicle.

$$\begin{aligned}\text{gradient} &= \frac{AB}{CB} \\ &= \frac{(200 - 100) \text{ m s}^{-1}}{(6.0 - 3.0) \text{ s}} \\ &= \frac{100 \text{ m s}^{-1}}{3.0 \text{ s}} \\ &= 33.3 \text{ m s}^{-2} \\ \therefore \text{acceleration} &= 33 \text{ m s}^{-2}\end{aligned}$$

Data	
Time (s)	Velocity (m s^{-1})
1.0	30
2.0	65
3.0	100
4.0	130
5.0	165
6.0	200

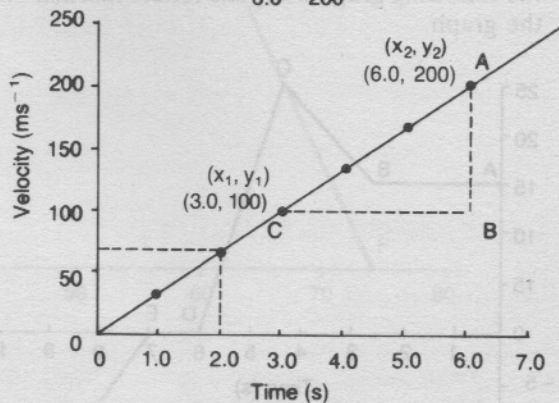


Figure 3.13

The area under the curve will give the displacement of the vehicle at any time. For the displacement after 2.0 s, the area under the curve is determined as follows:

$$\begin{aligned}\text{area under curve} &= \text{area of triangle} \\ &= \frac{1}{2} (\text{time} \times \text{velocity}) \\ &= \frac{1}{2} (2.0 \text{ s} \times 65 \text{ m s}^{-1}) \\ \therefore \text{displacement} &= 65 \text{ m}\end{aligned}$$

It should be remembered that the vehicle started from rest and increased its velocity to 65 m s^{-1} in 2.0 s. This means that the vehicle had an average velocity of 32.5 m s^{-1} over the 2.0 s period. The resulting displacement will then be as calculated by the area under the curve.

$$2 \times 32.5 = 65 \text{ m}$$

Review exercise

1 Graph the following data:

Displacement (m)	Time (s)
30	1.0
100	2.0
230	3.0
400	4.0
630	5.0

- a Determine the gradient of the curve at $t = 2.0$ s and $t = 5.0$ s. What does the gradient of this curve represent?
- b What type of motion does the curve represent?
- c What is the mathematical relationship represented by this graph?

- 2 Describe the motion of the vehicle depicted in the following graph using the letters marked on the graph.

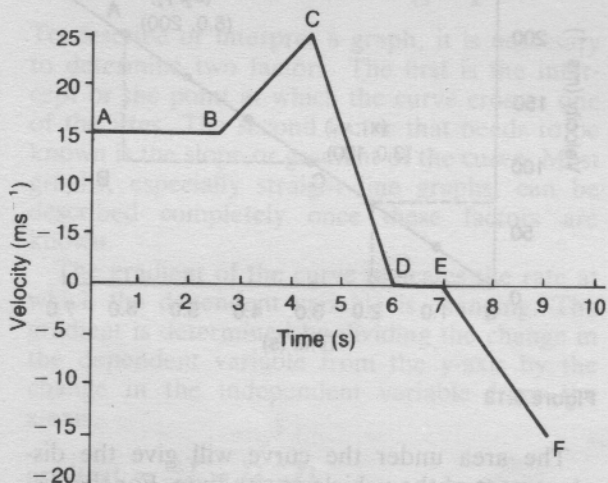


Figure 3.14

Major ideas

- Graphs can be used to illustrate the relationship between variables in physical relationships.
- The shape of a graph indicates whether there is a linear, inverse, parabolic, exponential or other relationship between the variables being graphed.
- The slope or gradient of a line graph indicates the rate of change between the variables being graphed.
- The area under the curve of a graph will give the product of the variables at any given point.

Revision questions

- 1 Would you expect the line of a graph to always pass through every plotted point? Explain your answer.
- 2 Explain what is meant by determining the 'line

of best fit' for a series of points on a graph.

- 3 Suggest how you could draw a graph to indicate the range of uncertainty for plotted values.
- 4 How could you determine the acceleration from a velocity-time graph?
- 5 Graph the following data.

x	y
1	36
2	67
3	93
4	142
5	280
6	483
7	1783

- a Describe the relationship between x and y.
- b How could you display this data to produce a straight line graph? Draw the graph to verify your answer.
- 6 Graph the following information.

Pressure (cm of Hg)	Volume (litres)
20	10.0
30	6.7
40	5.0
50	4.0
60	3.3
70	2.9

- a Describe the relationship between pressure and volume.
- b Graph this data to produce a straight line graph.
- 7 Describe the motion depicted in the graphs in Figure 3.15 and Figure 3.16.

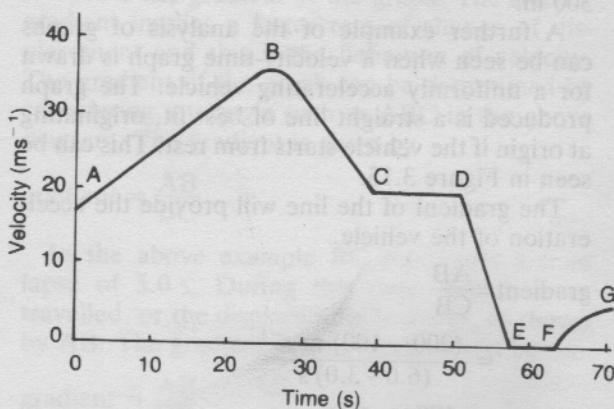


Figure 3.15